# Mathematical Comparison and Optimization of Urban Cycling Networks via Safety-Aware Clustering

Manuel Campero Jurado[1], Carlos Canudas-de-Wit[1], Giovanni De Nunzio[2]

***Abstract*—Comparing cycling networks across cities is challenging because networks differ in size, resolution, topology, and safety characteristics. Building on an existing exposure-based optimization model for cycling-infrastructure upgrades, this paper proposes a safety-aware framework for clustering, comparing, and transferring intervention strategies across urban cycling networks. Each network is represented as a weighted directed graph whose arcs encode degree of separation, distance, cyclist flow, traffic exposure, slope, and angular change. We introduce a constrained clustering method that reduces large networks to a common resolution while preserving safety-relevant quantities such as exposure, flow, and DoS consistency. This enables fair cross-city comparison using an exposure-based Gromov–Wasserstein distance, and allows clustered optimization solutions to be lifted back to the original network. We further use the Gromov–Wasserstein correspondence to transfer optimized strategies from a source city to structurally similar target cities. A case study on Grenoble, Strasbourg, and Lyon shows that the clustered representation preserves optimization quality while reducing runtime, and that GW similarity provides an interpretable signal for cross-city transferability of cycling-infrastructure interventions.**

***Index Terms*—Cycling networks, constrained clustering, cyclist exposure, bi-level optimization, urban mobility**

## I. Introduction

The health, ecological, and socio-economic benefits of a cycling society are well-documented, including improved cardiovascular health, increased longevity, reduced carbon footprint and air pollution, lower transportation costs, reduced traffic congestion, and greater convenience and efficiency [1], [2]. As a result, public authorities have made considerable efforts to promote cycling as an alternative to driving for short and medium distances. However, cyclists still face safety and comfort issues that may discourage them from riding [3]. It is well-established that infrastructure separating bicycles from motorized vehicles improves perceived safety and comfort, thereby encouraging higher cycling uptake [4], [5]. High-quality, well-connected cycling infrastructure can substantially increase ridership [6], [7]. Yet, analyzing real-world cycling networks remains challenging due to their size, spatial complexity, and the many interacting factors that influence route choice and perceived safety. In addition, being able to compare the safety performance of different urban cycling networks in a fair and scale-independent way would be highly valuable for planners seeking to replicate successful strategies from cities with more advanced infrastructure.

[1]Univ. Grenoble Alpes, Inria, CNRS, Grenoble INP, GIPSA-Lab, 38000 Grenoble, France {*manuel.campero-jurado@inria.fr, carlos.canudas-de-wit@gipsa-lab.fr*}

[2]IFP Energies nouvelles, Rond-point de l'échangeur de Solaize, BP 3, 69360 Solaize, France, giovanni.de-nunzio@ifpen.fr

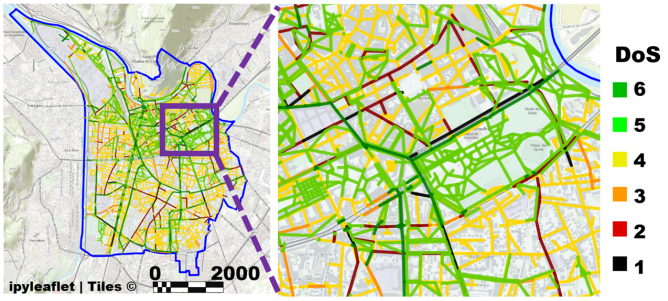



Fig. 1. Cycling network of Grenoble, France. The degree of separation (DoS) score of the cycling infrastructure from motorized vehicles is indicated in ascending order from 1 to 6.

We propose a mathematical framework to fairly compare cycling-network safety across cities of different sizes and detail levels. Large road graphs are compressed into cluster-based clustered-networks using a constrained, safety-aware k-means on a spectral embedding, preserving key attributes such as distance, slope, degree of separation, traffic factor, flow, and exposure while limiting information loss. On these clustered networks, we build exposure-based kernels to compare cities via structural similarity measures, and show on real cases that the clustered representations retain safety-critical structure well enough to enable consistent, interpretable cross-city safety comparisons.

Related work: Our work is based on two main areas: transport network clustering or community detection and network comparison.

*Community detection:* Cluster analysis and community detection are core tools for uncovering structure in complex data and networks. Classical $k$-means [8] partitions points into $K$ groups that minimize within-cluster variance, while network-specific methods identify densely connected node groups. Key approaches include edge-betweenness divisive methods [9], greedy modularity maximization for large graphs [10], spectral algorithms based on the modularity matrix [11], and fast physics-inspired methods using voltage-drop analogies [12].

These generic techniques are widely used in transportation. In traffic-flow analysis, $k$-means has been applied to define flow phases from detector data [13], detect disruptions on arterial networks [14], and characterize driving styles from automotive traces [15]. Hierarchical clustering has been used to derive recurrent daily profiles and distinguish workdays from weekends [16]. For spatially embedded transport networks, community detection supports homogeneous control regions and equity analysis: [17] use Infomap-based communities refined for homogeneity, contiguity, and compactness

in perimeter control. In the cycling context, [18] identify demand-based communities from taxi trips to design non-dominated bike networks, while [19] propose data-driven, budget-aware strategies to connect fragmented bicycle networks. Bike-and-ride network design models further show how optimization can shift trips from private cars to bicycle–transit combinations [20]. Compared with these works, our approach clusters cycling networks under explicit safety constraints so that the resulting clustered-networks preserve multi-attribute safety information and remain comparable across cities. To the best of our knowledge, existing clustering and community-detection methods are not explicitly designed to preserve cyclist-oriented safety indicators (e.g., exposure, slope, traffic conditions, angular change) when compressing large cycling networks for cross-city comparison.

*Network comparison:* Comparing two transport networks can be seen as a relaxation of graph isomorphism: instead of asking whether a bijection makes their adjacencies identical, we ask how far apart their structures are under a chosen distance [21]. A simple approach is to compare adjacency matrices via standard norms (Euclidean, Manhattan, etc.), in line with classical data analysis and data mining ideas [22], but such measures are highly sensitive to node labelling and local perturbations.

More robust comparisons use structural descriptors. Global statistics such as clustering coefficient, diameter, and average path length provide coarse summaries, while higher-order pattern counts (graphlets) and role-based features capture finer functional similarity across complex networks [23], [24]. Spectral methods use eigenvalues of adjacency or Laplacian matrices as label-invariant signatures, with spectral distances tracking graph edit distance in many cases [25]. From a more analytical viewpoint, cut distance and graphon theory offer limit objects and size-agnostic metrics with good empirical performance on diverse networks [26]. A complementary line embeds graphs into reproducing kernel Hilbert spaces via graph kernels. Early work defined positive definite kernels on labelled graphs despite the hardness of exact subgraph counting [27]. Random-walk and marginalized kernels [28], [29] exploit walk-based feature spaces while remaining computationally tractable, and are now standard tools for learning on graphs. All these approaches connect to optimal transport, which studies the most efficient way to redistribute mass. Wasserstein distances provide geometrically meaningful metrics between probability measures and have widespread applications in analysis and applied fields [30]–[32]. In our setting, this viewpoint underlies how we compare exposure distributions and safety-relevant kernels on cycling networks, yielding a principled notion of "distance" between cities that is sensitive to both weights and structure. However, existing methods are rarely tailored to safety-specific attributes of cycling networks, nor do they jointly handle multi-attribute exposure measures and differences in network size or resolution when drawing cross-city conclusions. To the best of the authors' knowlarc, there is currently no framework that (i) clusters cycling networks under explicit safety constraints so as to preserve multi-attribute indicators such as flow, slope, and angular change, and (ii) compares the resulting networks in a size-independent and mathematically consistent way using exposure-based kernels.

| DoS | Description | AADT (veh/day) |
|---|---|---|
| 6 | Dedicated bike lanes, fully separated from motor traffic. | 14 400 |
| 5 | Bike lanes shared with pedestrians, separate from cars. | 16 000 |
| 4 | Mixed-traffic roads (30 km/h) without designated bike lanes. | 960 |
| 3 | Shared roads with cycling pictograms. | 3 120 |
| 2 | Roads with painted bike lanes. | 6 912 |
| 1 | Unmarked shared car-bike lanes. | 8 448 |

TABLE I
THE DEGREE OF SEPARATION (DOS) CATEGORIZES BIKE LANE INFRASTRUCTURE BASED ON SAFETY LEVELS. THE THIRD COLUMN PRESENTS ANNUAL AVERAGE DAILY TRAFFIC (AADT) ON ADJACENT ROADS, SOURCED FROM THE GTLVILLE PLATFORM IN GRENOBLE [33].

*Statement of Contribution*: We propose a safety-aware framework for cross-city cycling-network analysis that: (i) models cyclist exposure using DoS, flow, traffic, distance, slope, and angular-change information; (ii) formulates original and clustered bi-level upgrade problems for reducing exposure under a budget; (iii) introduces a constrained clustering-and-lifting strategy that preserves safety-relevant network structure while reducing computational cost; and (iv) uses an exposure-based Gromov–Wasserstein (GW) distance to compare cities and transfer optimized infrastructure strategies between structurally similar networks. Experiments on Grenoble, Strasbourg, and Lyon show that the clustered approach closely approximates direct optimization and that GW similarity provides a useful basis for transferability.

## II. MODEL DESIGN

In this section, we first present the bike lane classification and the graph representation. We then define the cyclist flow model and the traffic-exposure measure on the original network. Next, we extend these quantities to the clustered representation, so that flow, exposure, and safety-relevant attributes are preserved after aggregation. Finally, we present the safety-aware clustering algorithm used to construct reduced networks that remain comparable across cities.

### A. Bike Lane Safety classification

The DoS classification, adapted from an OSM-based platform, categorizes cycling infrastructure based on separation from motor traffic. Table I presents DoS categories, showing that higher-DoS lanes offer greater safety. These scores serve as weights in the network graph, modeling upgrades from lower to higher safety levels. The classification methodology is detailed in [34]. Fig. 1 visualizes these classifications in the Grenoble cycling network.

### B. Graph Representation

Let the network be represented by a weighted directed graph (digraph) $\mathcal{G} = (\mathcal{V}, \mathcal{A}, \mathcal{W})$, where $\mathcal{V}$ is the set of nodes corresponding to intersections or road endpoints, $\mathcal{A}$ is the set of directed arcs representing roads, and $\mathcal{W}$ is the set of

weights assigned to those arcs. For each arc $a = (u, v) \in \mathcal{A}$, with $u, v \in \mathcal{V}$, the original weight $w_a^0 \in \mathcal{W}$ represents the DoS associated with the link. The weights are assumed to be symmetric, meaning $w_{(u,v)}^0 = w_{(v,u)}^0$. The geographical length of each arc $a$ is denoted by $d_a$.

### C. Cyclist Flows

Following [35], we use an all-or-nothing (AON) traffic assignment model that assumes no congestion and assigns all demand $q_{(u,v)}$ from the EMD2020 survey [36] to the shortest available path between nodes $u, v \in \mathcal{V}$. This reflects typical cycling dynamics, where congestion effects are minimal. In this section, we formalize and adopt the cyclist distance proposed in [37], which defines this distance using OpenStreetMap and Cardiff cycling flow data while incorporating infrastructure characteristics. Then, betweenness centrality is applied to estimate the flow distribution.

To account for cycling-specific factors, we use a modified distance metric that accounts for key factors influencing route choice. Let $\mathcal{V}$ denote the set of nodes in the network, and let $u, v \in \mathcal{V}$ be two nodes. The shortest path between $u$ and $v$ is represented as $p_{(u,v)} = (a_1, \ldots, a_k)$, where each arc $a \in p_{(u,v)}$ has a geographical length $d_a$, an angular change $\measuredangle_a$ (in degrees, either along $a$ or at junctions between consecutive arcs), a slope factor $s_a$ (which depends on the terrain slope $m_a$), and a traffic factor $f_a$ (that takes into account the flow of car traffic on roads adjacent to bike lanes). We define the cyclist distance $b_{(u,v)}$ between nodes $u$ and $v$ as

$$b_{(u,v)} = \sum_{a \in p_{(u,v)}} (d_a \cdot s_a \cdot (f_a)^\rho) + \sum_{a \in p_{(u,v)}} \measuredangle_a \cdot \eta, \quad (1)$$

where $\rho > 0$ and $\eta > 0$ are calibration parameters controlling the influence of traffic conditions and angular changes, respectively. The traffic factor $f_a$ is defined as

$$f_a(w_a^0) = \begin{cases} \psi \cdot \exp\left(\phi_a^{\text{AADT}}\right) & \text{if } w_a^0 < 5, \\ \psi \cdot 2.03 & \text{if } w_a^0 = 5, \\ \psi & \text{if } w_a^0 = 6. \end{cases} \quad (2)$$

Here, $\phi_a^{\text{AADT}}$ represents the annual average daily traffic (AADT) of the adjacent car lanes, expressed in thousands. The variable $w_a^0$ denotes the original DoS of the bike lane $a$. According to [38], the impact of annual average daily traffic (AADT) is neutralized when $w_a^0 \geq 5$, indicating full separation from car lanes. The factor 2.03 accounts for an observed increase in crash risk in shared spaces with pedestrians compared to separate, exclusive bicycle lanes [39]. The parameter $\psi$ is a scaling factor. Let $p_{(u,v)}$ be the shortest path between nodes $u$ and $v$ based on the cyclist distance in Eq. (1). Since the origin-destination (OD) demand in our data is symmetric, each node acts as both an origin and a destination, ensuring balanced inflow and outflow. The flow conservation for paths is given by

$$\phi_{p_{(u,v)}} = q_{(u,v)}, \quad (3)$$

where $q_{(u,v)}$ represents the demand between nodes $u$ and $v$. Next, let $\mathbb{I}_{a \in p_{(u,v)}}$ be a binary variable indicating whether arc $a$ is part of the shortest path $p_{(u,v)}$. Then, the total cyclist flow on arc $a$, denoted as $\phi_a$, is

$$\begin{aligned} \phi_a &= \sum_{u,v \in \mathcal{V}} q_{(u,v)} \cdot \mathbb{I}_{a \in p_{(u,v)}} \geq 0, \quad \forall a \in \mathcal{A} \\ \Phi &= \sum_{a \in \mathcal{A}} \phi_a. \end{aligned} \quad (4)$$

This ensures that the total flow on each arc is the sum of all demands for trips passing through it, maintaining balance across the network due to the symmetry of the demand matrix. To assess cyclist exposure to traffic conditions, we introduce $\mathcal{E}$ as a general measure of exposure, defined for paths in the network. This metric helps evaluate network improvements. For a shortest path $p_{(u,v)} = (a_1, \ldots, a_k)$, cyclist exposure captures the combined effect of traffic-related risk and the distance of cyclist interaction with traffic, and is defined as

$$\mathcal{E}_{p_{(u,v)}} = \sum_{a \in p_{(u,v)}} d_a \cdot f_a \cdot \phi_a, \quad (5)$$

where $\phi_a, d_a$ and $f_a$ are the normalized arc flow, length and traffic factor, respectively. For a single arc $a \in \mathcal{A}$, cyclist exposure simplifies to

$$\mathcal{E}_a = d_a \cdot f_a \cdot \phi_a \in [0, 1]. \quad (6)$$

Then, for the original network the total cyclist exposure is:

$$\mathcal{E}^{\text{orig}} = \frac{1}{|\mathcal{A}|} \sum_{(v_i,v_j) \in \mathcal{A}} \phi_{(v_i,v_j)} \cdot f(w_{(v_i,v_j)}) \cdot d_{(v_i,v_j)}. \quad (7)$$

Having presented the graph representation, cyclist flow model, and exposure-based safety indicator, we now turn to the clustering algorithm. The purpose of this step is to construct a reduced network that preserves the safety-relevant attributes introduced above, including flow, traffic exposure, slope, angular change, and infrastructure quality. This clustered representation provides the basis for comparing and optimizing cycling networks across cities in a fair and scale-independent manner.

### D. Cluster Network Definition

Let $\mathcal{P} = \{C_1, \ldots, C_k\}$ be a partition of the node set $\mathcal{V}$ into $k$ non-empty clusters. We define the clustered network as $\tilde{\mathcal{G}} = (\tilde{\mathcal{V}}, \tilde{\mathcal{A}}, \tilde{\mathcal{W}})$, where each cluster becomes a clustered-node, i.e., $\tilde{\mathcal{V}} = \mathcal{P}$.

For any pair of clusters $(C_r, C_{r'})$, let

$$\mathcal{A}(C_r, C_{r'}) = \left\{(u, v) \in \mathcal{A} \mid u \in C_r,\ v \in C_{r'}\right\}$$

denote the set of original arcs connecting them. A clustered-arc $\tilde{a} = (C_r, C_{r'})$ is included in $\tilde{\mathcal{A}}$ whenever $\mathcal{A}(C_r, C_{r'}) \neq \varnothing$. For notational simplicity, intra-cluster arcs are denoted by

$$\mathcal{A}(C_r) := \mathcal{A}(C_r, C_r).$$

The clustered DoS weights are defined conservatively using the minimum DoS among the underlying arcs. Thus, for $r \neq r'$,

$$\tilde{w}_{(C_r,C_{r'})}^{\text{inter}} = \min_{a \in \mathcal{A}(C_r,C_{r'})} w_a, \quad (8)$$

while the intra-cluster DoS weight is

$$\tilde{w}^{\text{intra}}_{C_r} = \min_{a \in \mathcal{A}(C_r)} w_a. \tag{9}$$

These definitions preserve the highest-risk connection between or within clusters.

### *E. Clustered cyclist distance*

To aggregate arc-level attributes in the clustered cycling network, we use flow-weighted averages. For any arc attribute $x_a$ and any non-empty arc set $S \subseteq \mathcal{A}$, define

$$\langle x \rangle^{\phi}_{S} := \frac{\sum_{a \in S} x_a \cdot \phi_a}{\sum_{a \in S} \phi_a}, \tag{10}$$

whenever $\sum_{a \in S} \phi_a > 0$. This choice gives greater importance to arcs that carry larger cyclist flows.

Let $\tilde{p}_{(C_r, C_{r'})} = (\tilde{a}_1, \ldots, \tilde{a}_m)$ denote a shortest path between clusters $C_r$ and $C_{r'}$ in $\tilde{\mathcal{G}}$, and let $\mathcal{C}(C_r, C_{r'}) \subseteq \tilde{\mathcal{V}}$ be the set of clusters visited along this path. The clustered cyclist distance is defined as

$$\tilde{b}_{(C_r, C_{r'})} = \sum_{\tilde{a} \in \tilde{p}_{(C_r, C_{r'})}} \tilde{b}^{\text{inter}}_{\tilde{a}} + \sum_{C_q \in \mathcal{C}(C_r, C_{r'})} \tilde{b}^{\text{intra}}_{C_q}. \tag{11}$$

For an inter-cluster clustered-arc $\tilde{a} = (C_r, C_{r'})$, with $r \neq r'$, we set

$$\tilde{b}^{\text{inter}}_{\tilde{a}} = \tilde{d}^{\text{inter}}_{\tilde{a}} \cdot \tilde{s}^{\text{inter}}_{\tilde{a}} \cdot \big(\tilde{f}^{\text{inter}}_{\tilde{a}}\big)^{\rho} + \tilde{\measuredangle}^{\text{inter}}_{\tilde{a}} \cdot \eta, \tag{12}$$

where

$$\begin{aligned} \tilde{d}^{\text{inter}}_{\tilde{a}} &= \langle d \rangle^{\phi}_{\mathcal{A}(C_r, C_{r'})}, & \tilde{s}^{\text{inter}}_{\tilde{a}} &= \langle s \rangle^{\phi}_{\mathcal{A}(C_r, C_{r'})}, \\ \tilde{\measuredangle}^{\text{inter}}_{\tilde{a}} &= \langle \measuredangle \rangle^{\phi}_{\mathcal{A}(C_r, C_{r'})}, & \tilde{f}^{\text{inter}}_{\tilde{a}} &= f\big(\tilde{w}^{\text{inter}}_{(C_r, C_{r'})}\big). \end{aligned} \tag{13}$$

Similarly, the intra-cluster contribution of cluster $C_r$ is

$$\tilde{b}^{\text{intra}}_{C_r} = \tilde{d}^{\text{intra}}_{C_r} \cdot \tilde{s}^{\text{intra}}_{C_r} \cdot \big(\tilde{f}^{\text{intra}}_{C_r}\big)^{\rho} + \tilde{\measuredangle}^{\text{intra}}_{C_r} \cdot \eta, \tag{14}$$

with

$$\begin{aligned} \tilde{d}^{\text{intra}}_{C_r} &= \langle d \rangle^{\phi}_{\mathcal{A}(C_r)}, & \tilde{s}^{\text{intra}}_{C_r} &= \langle s \rangle^{\phi}_{\mathcal{A}(C_r)}, \\ \tilde{\measuredangle}^{\text{intra}}_{C_r} &= \langle \measuredangle \rangle^{\phi}_{\mathcal{A}(C_r)}, & \tilde{f}^{\text{intra}}_{C_r} &= f\big(\tilde{w}^{\text{intra}}_{C_r}\big). \end{aligned} \tag{15}$$

### *F. Clustered demand, flow assignment, and exposure*

Given the original network $\mathcal{G}$ and its clustered representation $\tilde{\mathcal{G}}$, the clustered demand between two clusters is obtained by aggregating all original OD demands with origins in the first cluster and destinations in the second:

$$\tilde{q}_{(C_r, C_{r'})} := \sum_{\substack{u \in C_r \\ v \in C_{r'}}} q_{(u,v)}. \tag{16}$$

For the all-or-nothing assignment on the clustered graph, let $\tilde{p}_{(C_r, C_{r'})}$ be the shortest path between $C_r$ and $C_{r'}$ computed using the clustered cyclist distance in (11). The inter-cluster flow on a clustered-arc $\tilde{a} \in \tilde{\mathcal{A}}$ is

$$\tilde{\phi}^{\text{inter}}_{\tilde{a}} := \sum_{r, r'} \tilde{q}_{(C_r, C_{r'})} \cdot \mathbb{I}_{\tilde{a} \in \tilde{p}_{(C_r, C_{r'})}}. \tag{17}$$

Similarly, the intra-cluster flow associated with $C_q$ is the total clustered demand whose shortest path visits that cluster:

$$\tilde{\phi}^{\text{intra}}_{C_q} := \sum_{r, r'} \tilde{q}_{(C_r, C_{r'})} \cdot \mathbb{I}_{C_q \in \mathcal{C}(C_r, C_{r'})}. \tag{18}$$

The clustered total exposure is then defined as the average exposure over inter-cluster clustered-arcs and intra-cluster components:

$$\mathcal{E}^{\text{clust}} := \frac{\mathcal{E}^{\text{inter}} + \mathcal{E}^{\text{intra}}}{|\tilde{\mathcal{A}}| + k}, \tag{19}$$

where

$$\begin{aligned} \mathcal{E}^{\text{inter}} &:= \sum_{\tilde{a} \in \tilde{\mathcal{A}}} \tilde{\phi}^{\text{inter}}_{\tilde{a}} \cdot \tilde{f}^{\text{inter}}_{\tilde{a}} \cdot \tilde{d}^{\text{inter}}_{\tilde{a}}, \\ \mathcal{E}^{\text{intra}} &:= \sum_{r=1}^{k} \tilde{\phi}^{\text{intra}}_{C_r} \cdot \tilde{f}^{\text{intra}}_{C_r} \cdot \tilde{d}^{\text{intra}}_{C_r}. \end{aligned} \tag{20}$$

The corresponding total clustered flow is

$$\Phi^{\text{clust}} := \sum_{\tilde{a} \in \tilde{\mathcal{A}}} \tilde{\phi}^{\text{inter}}_{\tilde{a}} + \sum_{r=1}^{k} \tilde{\phi}^{\text{intra}}_{C_r}. \tag{21}$$

### *G. Clustering Algorithm (CC)*

One of our objectives is to compare cycling networks of different sizes, then, we cluster the larger network until its number of clustered-nodes matches the number of nodes in the smaller network. The goal is to construct a reduced graph $\tilde{\mathcal{G}}$ that remains faithful to the original network in terms of safety-relevant attributes, including DoS, distance, slope, traffic factor, cyclist flow, and exposure.

We first define a safety-aware weighted adjacency matrix:

$$A_{ij} = \begin{cases} w^0_{ij} \cdot \exp\!\left(-\dfrac{d_{ij}}{\gamma}\right) \cdot \left(\dfrac{\phi_{ij}}{\max \phi}\right)^{\alpha}, & \text{if } (i,j) \in \mathcal{A}, \\ 0, & \text{otherwise,} \end{cases} \tag{22}$$

where $w^0_{ij}$ is the DoS value, $d_{ij}$ is the spatial distance, $\phi_{ij}$ is the cyclist flow on arc $(i,j)$, $\gamma > 0$ controls the effect of spatial proximity, and $\alpha \geq 0$ controls the importance of high-flow arcs. The corresponding normalized Laplacian is

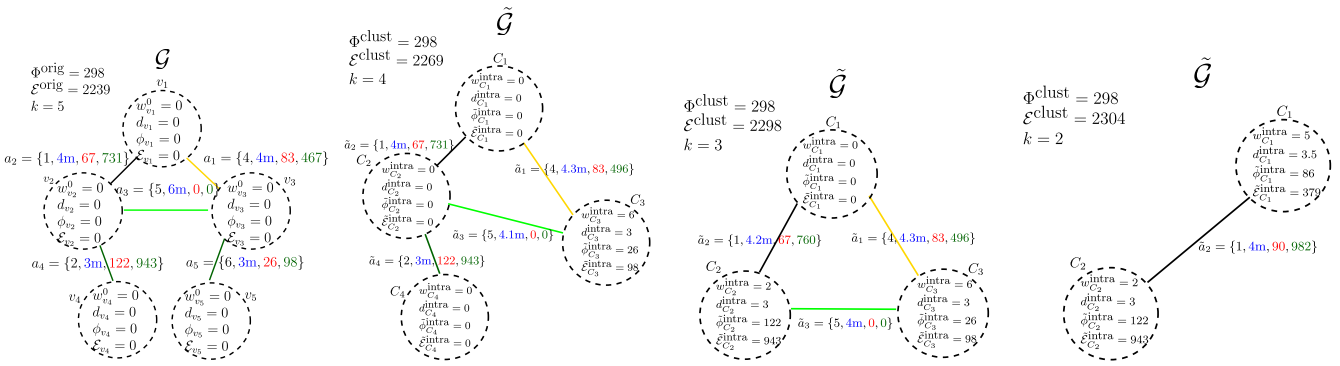
$$\mathcal{L} = I - D^{-1/2} A D^{-1/2}, \qquad D_{ii} = \sum_j A_{ij}. \tag{23}$$

A spectral embedding is obtained from $\mathcal{L}$, and clustering is then performed in this embedded space. The clustering step uses a constrained $k$-means procedure, summarized in Algorithm 1 and detailed in Appendix A and this clustered constrained algorithm will be called CC-Algorithm. The constraints are designed to preserve both local safety consistency and global network-level quantities. Specifically, the algorithm uses the following parameters:

$$\epsilon_{\max}, \quad \delta_{\max}, \quad \tau^{(0)}_{\mathcal{E}}, \quad \tau^{(0)}_{\Phi}, \quad \alpha_{\text{relax}}.$$

Here, $\epsilon_{\max}$ bounds the allowed DoS variation inside each cluster, while $\delta_{\max}$ limits the spatial diameter of a cluster. The initial tolerances

$$\tau^{(0)}_{\mathcal{E}} = \frac{\left|\mathcal{E}^{\text{clust}} - \mathcal{E}^{\text{orig}}\right|}{\mathcal{E}^{\text{orig}}}, \qquad \tau^{(0)}_{\Phi} = \frac{\left|\Phi^{\text{clust}} - \Phi^{\text{orig}}\right|}{\Phi^{\text{orig}}},$$




**(a)** Original **(b)** Cluster $k = 4$ **(c)** Cluster $k = 3$ **(d)** Cluster $k = 2$

Fig. 2. Algorithm 1 for toy example where $a = \{w_a^0, d_a(\text{m}), \phi_a, \mathcal{E}_a\}$, $\tilde{a}^{\text{inter}} = \left\{w_{\tilde{a}}^{\text{inter}}, d_{\tilde{a}}^{\text{inter}}(\text{m}), \phi_{\tilde{a}}^{\text{inter}}, \mathcal{E}_{\tilde{a}}^{\text{inter}}\right\}$, and $C^{\text{intra}} = \left\{d_C^{\text{intra}}, d_C^{\text{intra}}(\text{m}), \phi_C^{\text{inter}}, \mathcal{E}_C^{\text{intra}}\right\}$

control the admissible relative distortion in total exposure and total flow, respectively. The parameter $\alpha_{\text{relax}} > 1$ gradually relaxes these tolerances if no feasible clustering is found.

At each iteration, the algorithm applies $k$-means in the spectral embedding while penalizing violations of the exposure and flow constraints. DoS homogeneity and spatial compactness are enforced as hard constraints. If no feasible clustering is found, the exposure and flow tolerances are gradually relaxed; if the problem remains infeasible, the method falls back to standard spectral $k$-means. This ensures that a clustered network $\tilde{\mathcal{G}}$ is always obtained, while preserving the main safety and flow characteristics of the original graph. Fig. 2 illustrates the clustering procedure on a toy example. The constrained clustering procedure admits the following theoretical guarantees.

**Theorem** 1 (Legitimacy of the normalized Laplacian). *The matrix $\mathcal{L} = I - D^{-1/2}AD^{-1/2}$ is a valid symmetric normalized graph Laplacian.*

**Theorem** 2 (Flow Preservation in Network Clustering). *Given a di-graph $\mathcal{G} = (\mathcal{V}, \mathcal{A}, \mathcal{W})$ with demand matrix $q_{(u,v)}$ and its clustered representation $\tilde{\mathcal{G}} = (\tilde{\mathcal{V}}, \tilde{\mathcal{A}}, \tilde{\mathcal{W}})$ constructed via Algorithm 1, the total network flow is preserved:*

$$\Phi^{orig} \triangleq \sum_{a \in \mathcal{A}} \phi_a = \sum_{\tilde{a} \in \tilde{\mathcal{A}}} \tilde{\phi}_{\tilde{a}} = \Phi^{clust}.$$

*Proof.* The proofs are provided in Appendix A. □

We now formulate the infrastructure-upgrade problem. First, we present the original arc-level bi-level model, where upgrades are assigned to individual arcs. Then, we derive a clustered counterpart that reduces computational complexity while preserving the main exposure and routing effects.

## III. Optimization Framework

After defining the exposure measures and clustered representation, we formulate the infrastructure-upgrade problem. The goal is to decide which cycling links should be improved, and to what DoS level, under a fixed budget. Because upgrades affect route choice, we model the problem as a bi-level optimization: the upper level selects infrastructure upgrades, while the lower level recomputes cyclist flows through an all-or-nothing assignment. We first present the arc-level formulation and then introduce a clustered version that reduces computational complexity while preserving the interaction between upgrades, routing, and exposure.

### A. Original Arc-Level Optimization Problem

Given a fixed budget $B$, we model infrastructure upgrading as a bi-level optimization problem as in [40]. The upper level $\mathbb{U}_{\text{pper}}$ selects which arcs to upgrade and to what DoS level, with the goal of minimizing total cyclist exposure under the budget constraint. The lower level $\mathbb{L}_{\text{ower}}$ then recomputes cyclists' route choices using the AON assignment with the upgraded cyclist distances. Since DoS changes affect traffic factors, shortest paths, and flows, the two levels are coupled.

We define the following notation:

- $\mathbf{w}^0 = \left[w_a^0\right]_{a \in \mathcal{A}}^\top \in \mathbb{N}^{|\mathcal{A}|}$: vector of original DoS values for each arc in the bike network;
- $\mathbf{w} = [w_a]_{a \in \mathcal{A}}^\top \in \Omega$: vector of decision variables (candidate configuration) representing the upgraded DoS values, where $\Omega$ is the constraint set that defined as $\Omega = \left\{\mathbf{w} \in \mathbb{N}^{|\mathcal{A}|} : w_a^0 \le w_a \le 6, \quad \forall a \in \mathcal{A}\right\}$;
- $\pi(w_a^0, w_a) \in \mathbb{R}_{\ge 0}$: the cost or price per meter of upgrading arc $a$ from $w_a^0$ to $w_a$.

The original arc-level bi-level optimization problem is then written as follows:

**Problem 1.** *Original bi-level network-upgrade optimization*

*problem*

$\mathbb{U}_{\text{pper}}$ :

$$\mathbf{w}^\star = \arg\min_{\mathbf{w}\in\Omega} \frac{1}{|\mathcal{A}|}\sum_{a\in\mathcal{A}} \phi_a^\star(\mathbf{w})\cdot f_a(w_a)\cdot d_a \tag{24}$$

$$\text{s.t.} \sum_{a\in\mathcal{A}} d_a\cdot\pi(w_a^0, w_a) \le B, \tag{25}$$

$\mathbb{L}_{\text{ower}}$ :

$$\boldsymbol{\phi}^\star(\boldsymbol{w}) = \arg\min_{\boldsymbol{\phi}\in\mathbb{R}_{\ge 0}^{|\mathcal{A}|}} \sum_{a\in\mathcal{A}} \bar{d}_a(\mathbf{w})\cdot\phi_a, \tag{26}$$

$$\text{s.t. } \phi_a = \sum_{(u,v)\in\mathcal{V}_\mathcal{O}\times\mathcal{V}_\mathcal{D}} q_{(u,v)}\cdot\mathbb{I}_{a\in p_{(u,v)}(\mathbf{w})} \ge 0, \quad \forall a\in\mathcal{A}.$$

Then $\mathbf{w}^\star$ denotes the solution of Problem 1.

### *B. Cluster-Level Optimization*

Solving the original bi-level problem directly on $\mathcal{G}$ can be computationally demanding, since the upper level contains one decision variable for each arc in $\mathcal{A}$. We therefore formulate a clustered counterpart on the aggregated network $\tilde{\mathcal{G}}$, defined in the previous section.

The clustered model makes upgrade decisions on aggregated objects rather than on individual arcs. Let

$$\mathcal{G}_{\text{agg}} = \tilde{\mathcal{V}} \cup \tilde{\mathcal{A}}$$

denote the set of upgradeable objects, where elements of $\tilde{\mathcal{V}}$ represent intra-cluster infrastructure and elements of $\tilde{\mathcal{A}}$ represent inter-cluster connections. Thus, upgrading a cluster $C_r$ applies to the original arcs in $\mathcal{A}(C_r)$, while upgrading a clustered-arc $(C_r, C_s)$ applies to the original arcs in $\mathcal{A}(C_r, C_s)$.

For each object $g \in \mathcal{G}_{\text{agg}}$, let $w_g^0$ be its initial DoS level, defined using the conservative clustered DoS weights:

$$w_g^0 = \begin{cases} \tilde{w}^{\text{intra}}_{C_r}, & \text{if } g = C_r \in \tilde{\mathcal{V}}, \\ \tilde{w}^{\text{inter}}_{(C_r,C_s)}, & \text{if } g = (C_r, C_s) \in \tilde{\mathcal{A}}. \end{cases}$$

The clustered upgrade decision is $\tilde{\mathbf{w}} = [\tilde{w}_g]_{g\in\mathcal{G}_{\text{agg}}} \in \mathbb{N}^{|\mathcal{G}_{\text{agg}}|}$, where $\tilde{w}_g$ is the upgraded DoS level of object $g$. The feasible set is

$$\tilde{\Omega} = \left\{ \tilde{\mathbf{w}} \in \mathbb{N}^{|\mathcal{G}_{\text{agg}}|} : w_g^0 \le \tilde{w}_g \le 6, \quad \forall g \in \mathcal{G}_{\text{agg}} \right\}. \tag{27}$$

As in the arc-level formulation, the upgraded DoS determines the traffic factor. For each $g \in \mathcal{G}_{\text{agg}}$, we define

$$f_g(\tilde{w}_g) = \begin{cases} \psi\cdot\exp(\phi_g^{\text{AADT}}), & \tilde{w}_g < 5, \\ 2.03\cdot\psi, & \tilde{w}_g = 5, \\ \psi, & \tilde{w}_g = 6, \end{cases} \tag{28}$$

where $\phi_g^{\text{AADT}}$ is the aggregated AADT value associated with object $g$.

Given an upgrade vector $\tilde{\mathbf{w}}$, the clustered cyclist distance is computed as in Eq. (11), replacing the baseline traffic factors by the upgraded factors $f_g(\tilde{w}_g)$. For each clustered OD pair $(C_r, C_s)$, let $\tilde{p}_{(C_r,C_s)}(\tilde{\mathbf{w}})$ denote the resulting shortest path in $\tilde{\mathcal{G}}$. Given the clustered demand $\tilde{q}_{(C_r,C_s)}$, the lower-level all-or-nothing assignment sends all demand between $C_r$ and $C_s$ to $\tilde{p}_{(C_r,C_s)}(\tilde{\mathbf{w}})$. The induced inter-cluster $\tilde{\phi}_{\tilde{a}}^{\text{inter}}(\tilde{\mathbf{w}})$ and intra-cluster $\tilde{\phi}_{C_q}^{\text{intra}}(\tilde{\mathbf{w}})$ flows are defined as Eq. (17)-(18), respectively. Using the aggregated distances already defined in the clustered cyclist-distance formulation, the total clustered exposure under $\tilde{\mathbf{w}}$ is denoted by $\mathcal{E}^{\text{clust}}(\tilde{\mathbf{w}})$ and defined as in Eq. (19).

For each clustered upgradeable object $g \in \mathcal{G}_{\text{agg}}$, let $\mathcal{A}(g)$ be the set of original arcs represented by $g$:

$$\mathcal{A}(g) = \begin{cases} \mathcal{A}(C_r), & \text{if } g = C_r \in \tilde{\mathcal{V}}, \\ \mathcal{A}(C_r, C_s), & \text{if } g = (C_r, C_s) \in \tilde{\mathcal{A}}. \end{cases}$$

Since $w_g^0$ is defined as the minimum DoS among the arcs in $\mathcal{A}(g)$, an upgraded value $\tilde{w}_g$ is interpreted as a minimum target DoS for all arcs represented by $g$. Hence, only arcs with $w_a^0 < \tilde{w}_g$ are upgraded, while arcs already satisfying $w_a^0 \ge \tilde{w}_g$ remain unchanged.

The cost of upgrading object $g$ is therefore

$$\Pi_g(\tilde{w}_g) = \sum_{a\in\mathcal{A}(g)} d_a \cdot \pi\left(w_a^0, \max\{w_a^0, \tilde{w}_g\}\right), \tag{29}$$

where $d_a$ is the arc length and $\pi(\cdot,\cdot)$ is the unit cost per meter of the corresponding DoS upgrade and arcs that already have DoS greater than or equal to $\tilde{w}_g$ are left unchanged. The clustered bi-level network-upgrade problem is then

**Problem 2.** *Clustered bi-level network-upgrade optimization problem*

$\mathbb{U}_{\text{pper}}$ :

$$\tilde{\mathbf{w}}^\star = \arg\min_{\tilde{\mathbf{w}}\in\tilde{\Omega}} \mathcal{E}^{\text{clust}}(\tilde{\mathbf{w}}) \tag{30}$$

$$\text{s.t.} \sum_{g\in\mathcal{G}_{\text{agg}}} \Pi_g(\tilde{w}_g) \le B. \tag{31}$$

$\mathbb{L}_{\text{ower}}$ :

$\tilde{\boldsymbol{\phi}}^\star(\tilde{\mathbf{w}})$ *satisfies the AON assignment induced by Eq.* (11).

Problem 2 is the clustered counterpart of Problem 1. It optimizes DoS upgrades on aggregated objects under a budget constraint, while recomputing cyclist flows on the clustered network. The solution $\tilde{\mathbf{w}}^\star$ can then be lifted back to the original graph, providing a computationally lighter approximation that still captures the interaction between upgrades, route choice, and exposure.

## IV. Network Comparison

The purpose of this work is to compare the safety structure of two cycling networks of different sizes. Let $\mathcal{H} = (\mathcal{V}^\mathcal{H}, \mathcal{A}^\mathcal{H}, \mathcal{W}^\mathcal{H})$ be the smaller network, and let $\mathcal{G}$ be the larger network. We cluster $\mathcal{G}$ into $\tilde{\mathcal{G}} = (\tilde{\mathcal{V}}^\mathcal{G}, \tilde{\mathcal{A}}^\mathcal{G}, \tilde{\mathcal{W}}^\mathcal{G})$, where $\tilde{\mathcal{V}}^\mathcal{G} = \mathcal{C} = \{C_1, \ldots, C_k\}$, such that

$$|\mathcal{V}(\mathcal{H})| = |\tilde{\mathcal{V}}^\mathcal{G}| = k.$$

This size matching allows the two networks to be compared at the same resolution. In what follows, superscripts $\mathcal{H}$ and $\tilde{\mathcal{G}}$ indicate the network to which each quantity belongs.

We compare the two networks using a Gromov–Wasserstein (GW) distance based on their internal exposure structure. For each directed arc $(i,j) \in \mathcal{A}(\mathcal{H})$, we define the arc exposure as

$$\mathcal{E}_{ij}^{\mathcal{H}} = \phi_{ij}^{\mathcal{H}} \cdot f_{ij}^{\mathcal{H}} \cdot d_{ij}^{\mathcal{H}}, \tag{32}$$

where $\phi_{ij}^{\mathcal{H}}$ is the cyclist flow, $f_{ij}^{\mathcal{H}}$ is the traffic factor, and $d_{ij}^{\mathcal{H}}$ is the arc length. The exposure mass assigned to node $v_i \in \mathcal{V}(\mathcal{H})$ is the total exposure of its incident arcs:

$$\mu^{\mathcal{H}}(v_i) = \sum_{(i,j)\in\mathcal{A}(\mathcal{H})} \mathcal{E}_{ij}^{\mathcal{H}} + \sum_{(j,i)\in\mathcal{A}(\mathcal{H})} \mathcal{E}_{ji}^{\mathcal{H}}. \tag{33}$$

Analogously, for each super-arc $(C_r, C_s) \in \tilde{\mathcal{A}}^{\mathcal{G}}$, we define

$$\tilde{\mathcal{E}}_{rs}^{\tilde{\mathcal{G}}} = \tilde{\phi}_{rs}^{\text{inter}} \cdot \tilde{f}_{rs}^{\text{inter}} \cdot \tilde{d}_{rs}^{\text{inter}}, \tag{34}$$

where $\tilde{\phi}_{rs}^{\text{inter}}$ is the inter-cluster flow, $\tilde{f}_{rs}^{\text{inter}}$ is the inter-cluster traffic factor, and $\tilde{d}_{rs}^{\text{inter}}$ is the flow-weighted inter-cluster distance. The exposure mass of a clustered node $C_r \in \tilde{\mathcal{V}}^{\mathcal{G}}$ is defined by adding incoming, outgoing, and intra-cluster exposure:

$$\mu^{\tilde{\mathcal{G}}}(C_r) = \sum_{(C_r,C_s)\in\tilde{\mathcal{A}}^{\mathcal{G}}} \tilde{\mathcal{E}}_{rs}^{\tilde{\mathcal{G}}} + \sum_{(C_s,C_r)\in\tilde{\mathcal{A}}^{\mathcal{G}}} \left(\tilde{\mathcal{E}}_{sr}^{\tilde{\mathcal{G}}} + \tilde{\mathcal{E}}_{C_r}^{\text{intra}}\right). \tag{35}$$

The node masses are normalized into probability measures:

$$\hat{\mu}^{\mathcal{H}}(v_i) = \frac{\mu^{\mathcal{H}}(v_i)}{\sum_{u\in\mathcal{V}(\mathcal{H})} \mu^{\mathcal{H}}(u)}, \quad \hat{\mu}^{\tilde{\mathcal{G}}}(C_r) = \frac{\mu^{\tilde{\mathcal{G}}}(C_r)}{\sum_{s=1}^{k} \mu^{\tilde{\mathcal{G}}}(C_s)}. \tag{36}$$

These probability measures describe how cyclist exposure is distributed across the nodes of each network. To compare the internal structure of these exposure distributions, we define pairwise dissimilarity matrices for both networks. Let $x_i \in \mathbb{R}^2$ be the geographical coordinates of node $v_i \in \mathcal{V}(\mathcal{H})$, and let $c_r \in \mathbb{R}^2$ be the centroid of cluster $C_r$. We set

$$D_{ij}^{\mathcal{H}} = \left|\hat{\mu}^{\mathcal{H}}(v_i) - \hat{\mu}^{\mathcal{H}}(v_j)\right| + \gamma\|x_i - x_j\|_2, \quad i,j = 1,\dots,k, \tag{37}$$

and

$$D_{rs}^{\tilde{\mathcal{G}}} = \left|\hat{\mu}^{\tilde{\mathcal{G}}}(C_r) - \hat{\mu}^{\tilde{\mathcal{G}}}(C_s)\right| + \gamma\|c_r - c_s\|_2, \quad r,s = 1,\dots,k. \tag{38}$$

Here, $\gamma > 0$ balances exposure dissimilarity and geographical separation.

Let $\mathrm{T}(\hat{\mu}^{\mathcal{H}}, \hat{\mu}^{\tilde{\mathcal{G}}})$ be the set of admissible transport couplings between the two probability measures:

$$\mathrm{T}(\hat{\mu}^{\mathcal{H}}, \hat{\mu}^{\tilde{\mathcal{G}}}) = \Big\{ \mathrm{t} \in \mathbb{R}_+^{k\times k} \;\Big|\; \sum_{s=1}^{k} \mathrm{t}_{is} = \hat{\mu}^{\mathcal{H}}(v_i), \quad i = 1,\dots,k, \quad \sum_{i=1}^{k} \mathrm{t}_{is} = \hat{\mu}^{\tilde{\mathcal{G}}}(C_s), \quad s = 1,\dots,k \Big\}. \tag{39}$$

The squared Gromov–Wasserstein distance is then

$$\mathrm{t}^\star(\mathcal{H}, \tilde{\mathcal{G}}) = \sqrt{\min_{\mathrm{t}\in\mathrm{T}(\hat{\mu}^{\mathcal{H}},\hat{\mu}^{\tilde{\mathcal{G}}})} \sum_{i,j=1}^{k} \sum_{r,s=1}^{k} \left(D_{ij}^{\mathcal{H}} - D_{rs}^{\tilde{\mathcal{G}}}\right)^2 \mathrm{t}_{ir}\mathrm{t}_{js}}. \tag{40}$$

A small value of $\mathrm{t}^\star(\mathcal{H}, \tilde{\mathcal{G}})$ indicates that the two networks have similar internal exposure structures. In particular, it means that high-exposure and spatially important regions in the smaller network can be matched to analogous regions in the clustered larger network, without requiring a predefined node-to-node correspondence.

## V. Optimization Transfer Through Network Comparison

The previous sections introduced the two components that enable cross-city transfer: a clustered representation that puts networks at a common resolution, and a GW distance that measures similarity between their exposure-weighted structures. We now use the corresponding GW transport plan to investigate whether an upgrade strategy optimized on a source network can be transferred to a target network. In this way, the comparison step is used not only to measure similarity, but also to guide infrastructure interventions across cities.

We consider three complementary steps. First, each network is optimized independently under a common relative budget. Second, for clustered networks, we compare clustered optimization with original arc-level optimization after lifting the clustered solution back to the original graph. Third, we transfer an optimized upgrade pattern from a source network to a target network using the GW transport plan.

### A. Clustered Versus Original Optimization

For a network $\mathcal{G}$ and its clustered representation $\tilde{\mathcal{G}}$, we compare the original arc-level optimization with the cluster-level approximation. The original strategy is obtained by solving Problem 1 on the full graph, yielding $\mathbf{w}^\star = [w_a^\star]_{a\in\mathcal{A}}^\top \in \mathbb{N}^{|\mathcal{A}|}$. The clustered strategy is obtained by solving Problem 2 on the aggregated graph, yielding $\tilde{\mathbf{w}}^\star = \left[\tilde{w}_g^\star\right]_{g\in\mathcal{G}_{\text{agg}}}^\top \in \mathbb{N}^{|\mathcal{G}_{\text{agg}}|}$.

Because the initial DoS of a clustered object $g$ is defined as the minimum DoS among its underlying arcs, the clustered decision $\tilde{w}_g^\star$ is interpreted as a minimum target safety level for all arcs represented by $g$. Therefore, when lifting the clustered solution back to the original graph, an arc is upgraded only if its original DoS is below the optimized clustered DoS. For every $a \in \mathcal{A}(g)$, we define

$$w_a^{\star,\text{lift}} = \max\left\{w_a, \tilde{w}_g^\star\right\}. \tag{41}$$

Thus, arcs that are less safe than the clustered target are upgraded to $\tilde{w}_g^\star$, while arcs that already have DoS greater than or equal to this value remain unchanged. The lifted vector is denoted by $\mathbf{w}^{\star,\text{lift}} = \left[w_a^{\star,\text{lift}}\right]_{a\in\mathcal{A}}^\top$. The lifted strategy is then evaluated on the original network by recomputing the cyclist distances, the all-or-nothing assignment, and the total exposure $\mathcal{E}^{\text{orig}}(\mathbf{w}^{\star,\text{lift}})$.

### B. GW-Based Transfer of Upgrade Strategies

We now use the GW correspondence to construct an approximate solution of Problem 1 on the target network $\mathcal{G}$. Instead of solving the full arc-level problem directly on $\mathcal{G}$, we transfer an optimized strategy from a source network $\mathcal{H}$ whose safety-exposure structure is similar to that of the clustered target

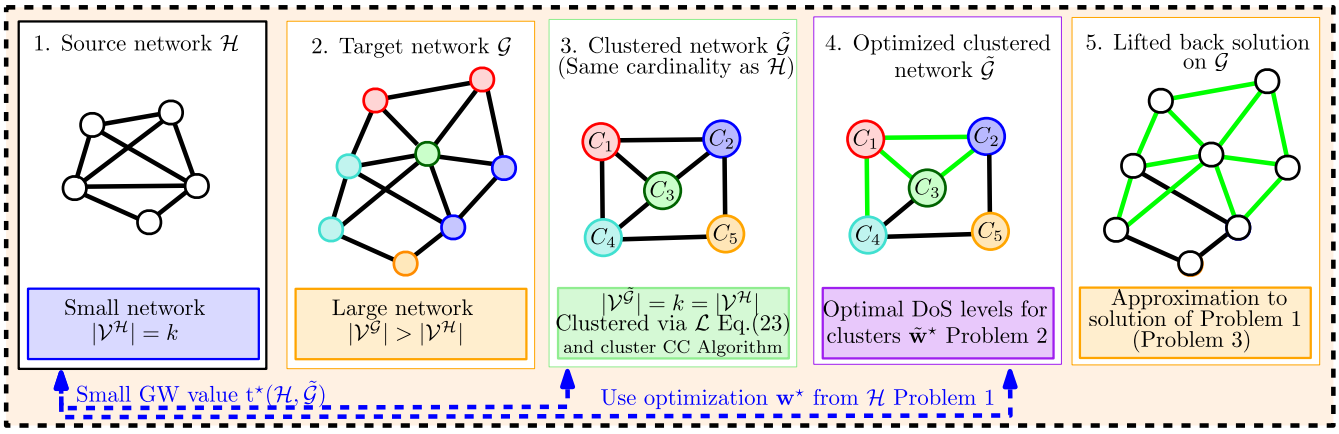



Fig. 3. Overview of the transfer framework: the target network $\mathcal{G}$ is clustered to match the source network $\mathcal{H}$; the optimized strategy from $\mathcal{H}$ is transferred to $\tilde{\mathcal{G}}$ and then lifted back to $\mathcal{G}$ as an approximate arc-level solution.

network $\tilde{\mathcal{G}}$. This similarity is measured by a small value of $GW(\mathcal{H}, \tilde{\mathcal{G}})$.

Following the clustering procedure introduced above, $\mathcal{G}$ is first reduced to

$$\tilde{\mathcal{G}} = (\tilde{\mathcal{V}}^{\mathcal{G}}, \tilde{\mathcal{A}}^{\mathcal{G}}, \tilde{\mathcal{W}}^{\mathcal{G}})$$

such that

$$|\tilde{\mathcal{V}}^{\mathcal{G}}| = |\mathcal{V}^{\mathcal{H}}|.$$

This gives both networks the same resolution. Let $\mathrm{t}^\star$ denote the optimal GW transport plan between $\mathcal{H}$ and $\tilde{\mathcal{G}}$, as defined in Section IV.

Assume that the source network $\mathcal{H}$, with baseline DoS vector $\mathbf{w}^{0,\mathcal{H}} = \left[w_a^{0,\mathcal{H}}\right]^\top_{a\in\mathcal{A}^{\mathcal{H}}} \in \mathbb{N}^{|\mathcal{A}^{\mathcal{H}}|}$, has been optimized under budget $B^{\mathcal{H}}$, yielding $\mathbf{w}^{\star,\mathcal{H}} = \left[w_a^{\star,\mathcal{H}}\right]^\top_{a\in\mathcal{A}^{\mathcal{H}}} \in \mathbb{N}^{|\mathcal{A}^{\mathcal{H}}|}$. We summarize the source upgrade around each node $v_i \in \mathcal{V}^{\mathcal{H}}$ by

$$u_i^{\mathcal{H}} = \frac{\sum_{a\in\mathcal{N}^{\mathcal{H}}(i)} \mathcal{E}_a^{\mathcal{H}}(\mathbf{w}^{\mathcal{H}}) \cdot \left(w_a^{\star,\mathcal{H}} - w_a^{0,\mathcal{H}}\right)}{\sum_{a\in\mathcal{N}^{\mathcal{H}}(i)} \mathcal{E}_a^{\mathcal{H}}(\mathbf{w}^{\mathcal{H}}) + \varepsilon}, \tag{42}$$

where $\mathcal{N}^{\mathcal{H}}(i)$ is the set of arcs incident to node $i$, $\mathcal{E}_a^{\mathcal{H}}(\mathbf{w}^{\mathcal{H}})$ is the baseline exposure of arc $a$, and $\varepsilon > 0$ avoids division by zero. Thus, $u_i^{\mathcal{H}}$ is large when highly exposed arcs around node $i$ receive large DoS improvements in the optimized source solution.

The GW plan $\mathrm{t}^\star$ transfers the source upgrade intensities to the clustered target network. For each cluster $C_r \in \tilde{\mathcal{V}}^{\mathcal{G}}$, we define

$$u_{C_r}^{\mathcal{H}\to\tilde{\mathcal{G}}} = \frac{\sum_i \mathrm{t}^\star_{ir} \cdot u_i^{\mathcal{H}}}{\sum_i \mathrm{t}^\star_{ir} + \varepsilon}. \tag{43}$$

The value $u_{C_r}^{\mathcal{H}\to\tilde{\mathcal{G}}}$ is the transferred upgrade score of cluster $C_r$. It is a weighted average of the source upgrade intensities $u_i^{\mathcal{H}}$, where the weights are given by the GW coupling $\mathrm{t}^\star_{ir}$. Therefore, this score is large when cluster $C_r$ is strongly matched to source nodes that were strongly upgraded in the optimized solution on $\mathcal{H}$.

We then assign a priority score $S_g$ to each upgradeable object $g \in \mathcal{G}^{\tilde{\mathcal{G}}}_{\mathrm{agg}}$. If $g$ is an intra-cluster object, its score is the transferred score of that cluster. If $g$ is an inter-cluster object, its score is the average transferred score of the two endpoint clusters:

$$S_g = \begin{cases} u_{C_r}^{\mathcal{H}\to\tilde{\mathcal{G}}}, & \text{if } g = C_r, \\ \dfrac{u_{C_r}^{\mathcal{H}\to\tilde{\mathcal{G}}} + u_{C_s}^{\mathcal{H}\to\tilde{\mathcal{G}}}}{2}, & \text{if } g = (C_r, C_s). \end{cases} \tag{44}$$

Thus, $S_g$ measures how important object $g$ is according to the upgrade pattern transferred from $\mathcal{H}$.

The transferred clustered strategy is obtained by maximizing the total priority of the selected upgrades under the target budget. Let

$$\tilde{\mathbf{w}} = [\tilde{w}_g]^\top_{g\in\mathcal{G}^{\tilde{\mathcal{G}}}_{\mathrm{agg}}} \in \tilde{\Omega}^{\tilde{\mathcal{G}}}$$

be a candidate DoS vector on the clustered target network. The objective rewards assigning higher DoS levels to clustered objects with larger transferred priority scores.

**Problem 3.** *GW-based transferred clustered upgrade problem*

$$\tilde{\mathbf{w}}^{\star,\mathrm{trans},\tilde{\mathcal{G}}} = \arg\max_{\tilde{\mathbf{w}}\in\tilde{\Omega}^{\tilde{\mathcal{G}}}} \sum_{g\in\mathcal{G}^{\tilde{\mathcal{G}}}_{\mathrm{agg}}} S_g \cdot \tilde{w}_g \tag{45}$$

$$\text{s.t.} \quad \sum_{g\in\mathcal{G}^{\tilde{\mathcal{G}}}_{\mathrm{agg}}} \Pi_g(\tilde{w}_g) \le B^{\mathcal{G}}. \tag{46}$$

Here, $S_g$ is the transferred priority score of object $g$, $\tilde{w}_g$ is the DoS assigned to $g$, $\Pi_g(\tilde{w}_g)$ is the aggregated cost of raising the original arcs represented by $g$ to at least $\tilde{w}_g$, and $B^{\mathcal{G}}$ is the available budget for the target network.

Problem 3 is a budgeted projection of the source solution onto $\tilde{\mathcal{G}}$. It does not solve the full bi-level Problem 1 on $\mathcal{G}$ directly. Instead, it uses the GW transport plan to identify which clustered objects of the target network should receive higher safety levels, based on the optimized intervention pattern of the source network, Fig. 3 shows a summary of the whole framework.

## VI. Solution Algorithms

The proposed problems are discrete bi-level network-design problems, where the upper level selects integer DoS upgrades

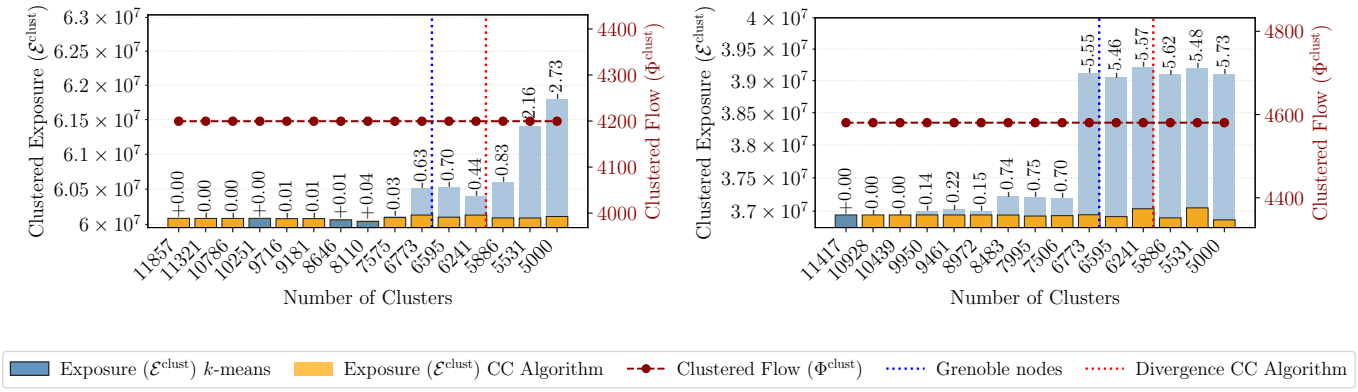



Fig. 4. Evolution of cyclist exposure and cyclist flow through the Algorithm A, at the left Lyon and on the right Strasbourg case.

and the lower level recomputes cyclist flows through AON assignment. Since related formulations are NP-hard [41], we use genetic algorithms to approximate both the arc-level and clustered problems. Each candidate DoS vector is evaluated by updating traffic factors, recomputing routes and flows, and using the resulting exposure as fitness; budget-infeasible candidates are repaired or penalized. The corresponding pseudo-codes are provided in Appendix B.

### A. GA for the Original Arc-Level Problem (Problem 1)

For Problem 1, each individual in the GA population is an arc-level DoS vector $\mathbf{w} = [w_a]_{a\in\mathcal{A}}^{\top} \in \Omega$. The fitness of a candidate solution is the total exposure in the upper-level objective of Problem 1, computed after solving the lower-level AON assignment induced by $\mathbf{w}$.

### B. GA for the Clustered Optimization Problem (Problem 2)

For Problem 2, the GA operates on aggregated objects rather than individual arcs. Each individual is a clustered DoS vector $\tilde{\mathbf{w}} = [\tilde{w}_g]_{g\in\mathcal{G}_{\text{agg}}}^{\top} \in \tilde{\Omega}$. The lower-level AON assignment is computed on the clustered network $\tilde{\mathcal{G}}$, and the fitness is the clustered exposure $\mathcal{E}^{\text{clust}}(\tilde{\mathbf{w}})$. The budget is evaluated using the aggregated upgrade cost $\Pi_g(\tilde{w}_g)$.

Once $\tilde{\mathbf{w}}^\star$ is obtained, it can be lifted back to the original graph. If an original arc $a$ belongs to the aggregated object $g$, then its lifted DoS is $w_a^{\star,\text{lift}} = \max\left\{w_a^0, \tilde{w}_g^\star\right\}$. This rule is consistent with the conservative clustered DoS definition: the optimized clustered value $\tilde{w}_g^\star$ represents a minimum target DoS for all original arcs represented by $g$, while arcs that are already safer are left unchanged.

### C. GW-Based Transfer Algorithm (Problem 3)

The third algorithm uses the GW correspondence to transfer an optimized strategy from a source network $\mathcal{H}$ to a target network $\mathcal{G}$. The target network is first clustered into $\tilde{\mathcal{G}}$ so that $|\tilde{\mathcal{V}}^{\mathcal{G}}| = |\mathcal{V}^{\mathcal{H}}|$. Then, the GW transport plan maps source upgrade intensities to clusters of $\tilde{\mathcal{G}}$. The transferred scores are used to prioritize clustered objects under the target budget, and the resulting strategy is lifted back to the original network $\mathcal{G}$.

Algorithms 2 and 3 approximate the solutions of the original and clustered bi-level optimization problems, respectively. Algorithm 4 provides a transfer-based alternative: it uses the optimized intervention pattern of a structurally similar source network to guide upgrades on the target network without solving the full arc-level problem directly.

The algorithms above provide the computational backbone of the proposed framework. The original GA gives a direct arc-level benchmark, the clustered GA tests whether optimization can be performed on reduced networks while preserving the main exposure effects, and the GW-transfer procedure evaluates whether an optimized strategy from one city can guide interventions in another. We now apply these three components to real cycling networks in order to assess their comparability, optimization performance, and transferability.

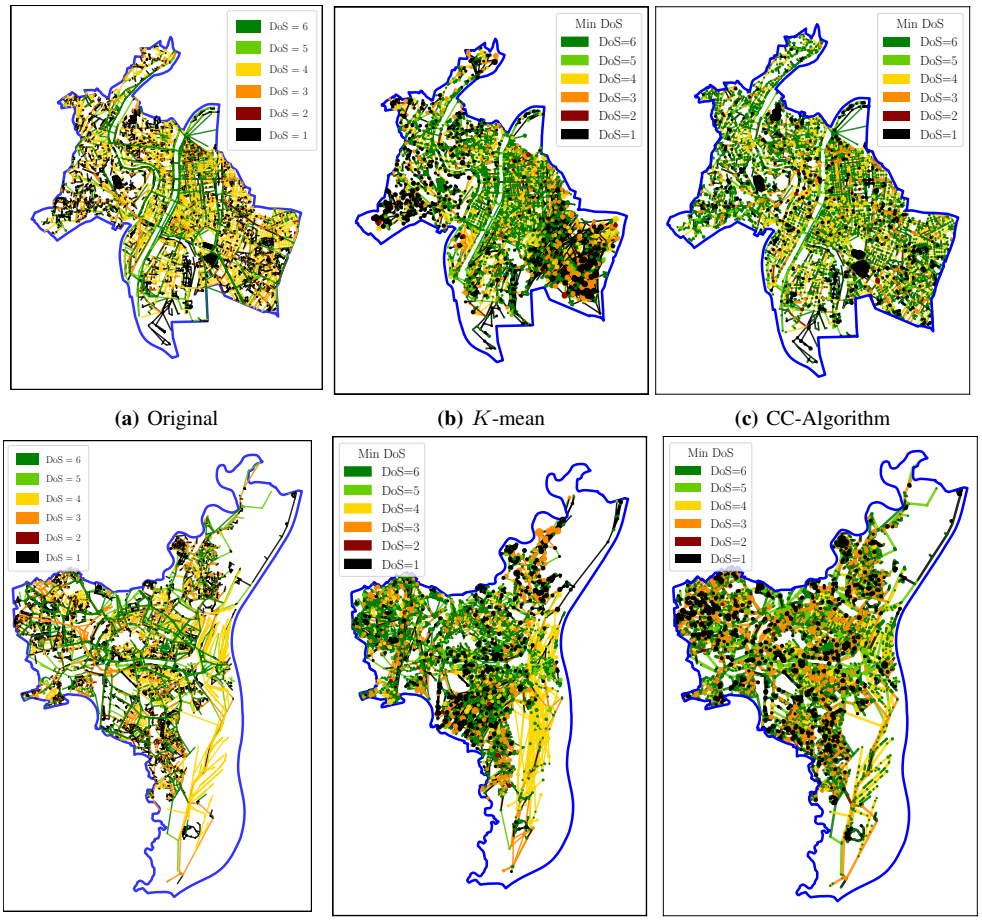
## VII. Case Study

We evaluate the proposed framework on three French cities with well-developed cycling networks: Lyon, Strasbourg, and Grenoble. The corresponding networks contain 11 857, 11 417, and 6 773 nodes, respectively. Since Grenoble is the smallest network, it is used as the reference network $\mathcal{H}^{\text{Grenoble}}$. Lyon and Strasbourg are clustered using CC-Algorithm to obtain reduced networks $\tilde{\mathcal{G}}^{\text{Lyon}}$ and $\tilde{\mathcal{G}}^{\text{Strasbourg}}$ with the same number of nodes as Grenoble:

$$|\tilde{\mathcal{V}}^{\text{Lyon}}| = |\tilde{\mathcal{V}}^{\text{Strasbourg}}| = |\mathcal{V}^{\text{Grenoble}}| = 6\,773.$$

This common resolution allows us to compare safety-exposure structures across cities using the GW distance and to test whether optimized intervention strategies can be transferred between structurally similar networks.

Fig. 4 evaluates the quality of the clustering step by tracking clustered exposure $\mathcal{E}^{\text{clust}}$ and clustered flow $\Phi^{\text{clust}}$ as the number of clusters decreases. Compared with unconstrained $k$-means, Algorithm 1 better preserves exposure and flow, especially under stronger compression. Strasbourg remains stable over a wider range of cluster counts, while Lyon begins to deviate earlier, suggesting that Strasbourg's safety-exposure structure is more easily compressed. The spatial comparison




**(a)** Original **(b)** $K$-mean **(c)** CC-Algorithm

Fig. 5. Clustering process with CC-Algorithm, on the first column we have the original networks, then the cluster one with $K$-mean and finally with CC-Algorithm.

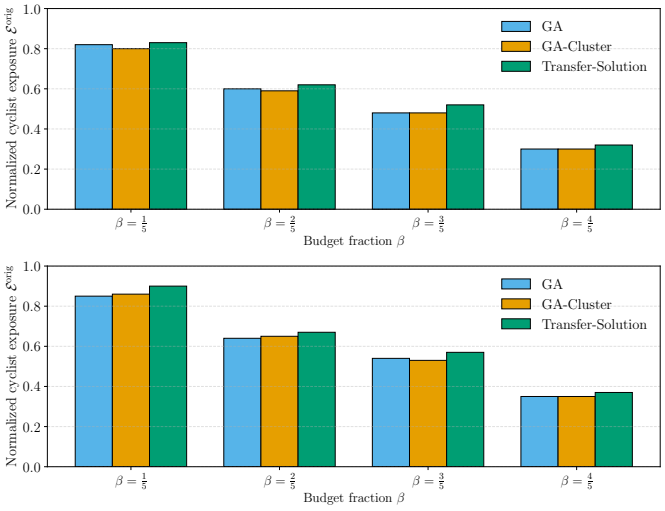



Fig. 6. Normalized cyclist traffic exposure under different optimization strategies: top, Strasbourg; bottom, Lyon.

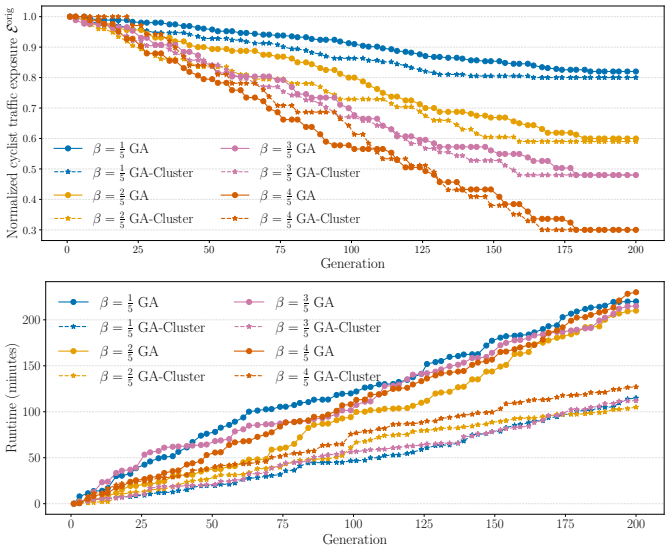
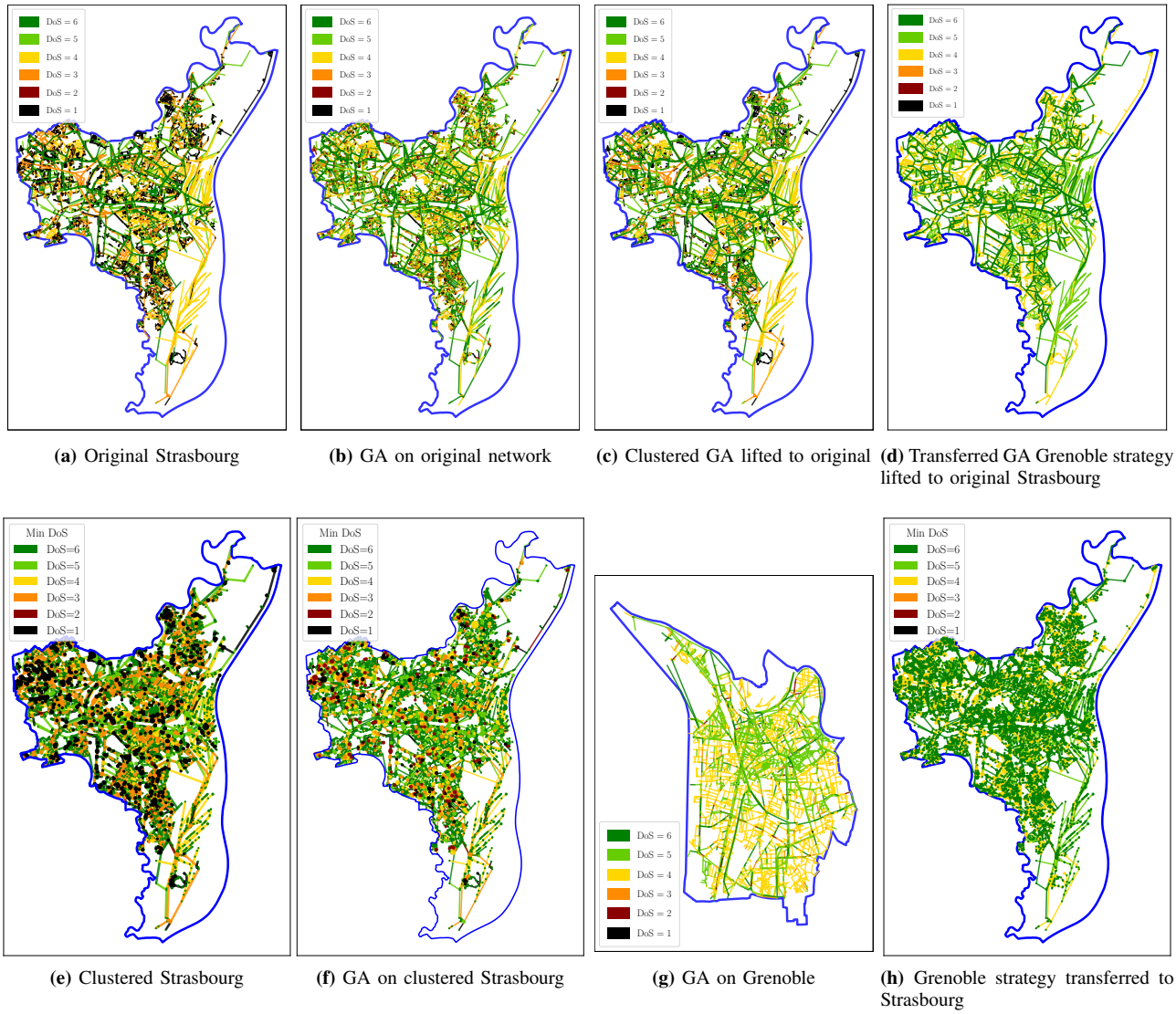



Fig. 7. Evolution of normalized cyclist traffic exposure and runtime over generations

in Fig. 5 supports this result: the constrained clustering better preserves high-risk corridors and low-risk greenways by enforcing DoS consistency and spatial compactness.

After clustering, we compare Lyon and Strasbourg with Grenoble using the GW distance introduced in Section IV.




**(a)** Original Strasbourg **(b)** GA on original network **(c)** Clustered GA lifted to original **(d)** Transferred GA Grenoble strategy lifted to original Strasbourg

**(e)** Clustered Strasbourg **(f)** GA on clustered Strasbourg **(g)** GA on Grenoble **(h)** Grenoble strategy transferred to Strasbourg

Fig. 8. Optimization and transfer strategies for the Strasbourg cycling network with $\beta = \frac{2}{5}$. The first row shows arc-level configurations on the original network: (a) initial DoS, (b) direct GA solution, (c) clustered GA solution lifted back to the original network, and (d) Grenoble strategy transferred through the GW correspondence and lifted to Strasbourg. The second row shows the corresponding clustered and source-network representations: (e) clustered Strasbourg, (f) GA solution on clustered Strasbourg, (g) GA solution on Grenoble, and (h) Grenoble strategy transferred to Strasbourg. Colors indicate DoS levels, with lower values corresponding to less separated cycling infrastructure.

The results indicate that Strasbourg is closer to Grenoble than Lyon is in terms of exposure-weighted network structure. This is consistent with the previous compression results and suggests that the optimized Grenoble strategy should be more transferable to Strasbourg. The ordering also agrees with external evidence from the Baromètre Vélo survey [42], where Grenoble and Strasbourg are typically ranked among the most bicycle-friendly French cities, while Lyon tends to score slightly lower.

We then evaluate three optimization strategies: direct arc-level GA optimization, clustered GA optimization followed by lifting, and GW-based transfer of the optimized Grenoble strategy. The available budget is defined as a fraction of the maximum budget required to upgrade all arcs to DoS 6:$B = \beta \cdot B_{\max}$, $B_{\max} = \sum_{a \in \mathcal{A}} d_a \cdot \pi(w_a^0, 6)$, with $\beta \in \left\{\frac{1}{5}, \frac{2}{5}, \frac{3}{5}, \frac{4}{5}\right\}$. Fig. 6 shows that the lifted clustered solution closely follows the direct GA solution for both Lyon and Strasbourg across the tested budget levels. This confirms that the clustered representation preserves the main information needed for optimization while reducing the search space. The transferred Grenoble solution also reduces exposure as the budget increases, with a smaller gap for Strasbourg, which is consistent with its stronger GW similarity to Grenoble.

Fig. 8 illustrates these effects on the Strasbourg network for $\beta = 2/5$. The direct GA and lifted clustered GA produce similar spatial intervention patterns on the original graph, showing that clustered optimization can be lifted back without losing the main structure of the solution. The transferred Grenoble strategy also upgrades several comparable high-priority corridors, although it differs from the direct Strasbourg solution because it is guided by the source network. This highlights the interpretability of both the clustering-and-lifting

strategy and the GW-based transfer mechanism.

Finally, Fig. 7 compares convergence and runtime. The lifted clustered GA reaches exposure levels close to the direct GA, often with faster convergence, and requires substantially less computational time. This demonstrates the practical benefit of solving the optimization problem on $\tilde{\mathcal{G}}$ and lifting the solution back to the original network. Overall, the case study shows that the proposed framework can compress large cycling networks while preserving safety-relevant structure, optimize infrastructure upgrades efficiently, and transfer intervention strategies across cities when their exposure-weighted structures are sufficiently similar.

## VIII. Strengths and Limitations

The proposed framework provides a unified approach to compare, compress, optimize, and transfer cycling-network intervention strategies using safety-relevant quantities such as DoS, cyclist flow, traffic exposure, slope, and angular change. Its main advantage is that the clustered representation reduces the optimization dimension while preserving the main exposure and flow structure of the original network. As a result, lifted clustered solutions can remain close to direct arc-level GA solutions while requiring lower computational cost. In addition, the GW-based comparison offers an interpretable measure of cross-city similarity and a principled basis for transferring optimized strategies between structurally similar networks.

The framework also has limitations. Its performance depends on the quality of the input data, including DoS classification, traffic estimates, OD demand, and slope information. The AON assignment is tractable but does not capture heterogeneous cyclist preferences, stochastic route choice, or behavioral adaptation after infrastructure changes. Moreover, clustering remains an approximation: even though the lifting step preserves a minimum target DoS for the original arcs, some local details inside clusters may be smoothed. Finally, the transfer procedure assumes that GW similarity is a useful proxy for intervention transferability, an assumption that should be further tested on a larger set of cities.

## IX. Conclusions and Future Work

This paper introduced a safety-aware framework for comparing and optimizing cycling networks across cities. The method combines exposure-based network metrics, constrained clustering, bi-level infrastructure optimization, and GW-based transfer of upgrade strategies. By clustering larger networks to the same resolution as a reference city, the framework enables fair cross-city comparison while reducing the optimization search space. The case study on Grenoble, Strasbourg, and Lyon shows that clustered optimization can be lifted back to the original network with competitive exposure reductions and lower runtime, while the transfer results suggest that cities with smaller GW distances share more compatible intervention patterns.

Future work will focus on richer route-choice models that capture heterogeneous and stochastic cyclist behavior, and on clustering constraints that incorporate equity, accessibility [43], and connectivity to key urban facilities [40]. The transfer methodology should also be tested on more cities to better identify when optimized strategies generalize across networks. Finally, the optimization model could be extended to multi-objective settings that jointly consider exposure reduction, cost, coverage, equity, and implementation feasibility.

## APPENDIX A

*Complementary Proofs*

*Proof of Theorem 1.* The adjacency matrix $A$ is symmetric because the arc attributes used to define it are symmetric, namely

$$w^0_{(u,v)} = w^0_{(v,u)}, \qquad d_{(u,v)} = d_{(v,u)}, \qquad \phi_{(u,v)} = \phi_{(v,u)}.$$

Moreover, all terms entering $A_{ij}$ are non-negative, and $A_{ij} = 0$ whenever $(i,j) \notin \mathcal{A}$. Hence, $A = A^\top$ and $A$ is non-negative.

Let $D$ be the diagonal degree matrix and $L = D - A$ the unnormalized Laplacian. For any $\mathbf{y} \in \mathbb{R}^{|\mathcal{V}|}$,

$$\mathbf{y}^\top L\mathbf{y} = \frac{1}{2}\sum_{i,j} A_{ij}(y_i - y_j)^2 \geq 0,$$

so $L$ is positive semidefinite. Since

$$\mathcal{L} = I - D^{-1/2}AD^{-1/2} = D^{-1/2}LD^{-1/2},$$

it follows that, for any $\mathbf{x}$,

$$\mathbf{x}^\top \mathcal{L}\mathbf{x} = (D^{-1/2}\mathbf{x})^\top L(D^{-1/2}\mathbf{x}) \geq 0.$$

Therefore, $\mathcal{L}$ is symmetric positive semidefinite and is a valid symmetric normalized graph Laplacian. □

*Proof of Theorem 2.* The clustered demand is defined by aggregating all OD demands between nodes belonging to the corresponding clusters:

$$\tilde{q}_{(C_r,C_{r'})} = \sum_{\substack{u\in C_r \\ v\in C_{r'}}} q_{(u,v)}.$$

Therefore, total demand is preserved:

$$\sum_{u,v\in\mathcal{V}} q_{(u,v)} = \sum_{C_r,C_{r'}\in\tilde{\mathcal{V}}} \tilde{q}_{(C_r,C_{r'})}.$$

The AON assignment sends each OD demand to exactly one shortest path in the original network and each clustered OD demand to exactly one shortest path in the clustered network. Hence, clustering redistributes the same total demand over aggregated paths, without creating or removing cyclist demand. Consequently, the total assigned flow mass is preserved:

$$\Phi^{\text{orig}} = \sum_{u,v\in\mathcal{V}} q_{(u,v)} = \sum_{C_r,C_{r'}\in\tilde{\mathcal{V}}} \tilde{q}_{(C_r,C_{r'})} = \Phi^{\text{clust}}.$$

□

## APPENDIX B

Now we present the Pseudo-Code of CC-Algorithm and Solution Algorithm for Problem 2-

**Algorithm 1** Constrained Spectral Clustering for Bike Networks (CC-Algorithm)

**Require:** $\mathcal{G} = (\mathcal{V}, \mathcal{A}, \mathcal{W})$, $k$, $\epsilon_{\max}$, $\delta_{\max}$, $\tau_{\mathcal{E}}^{(0)}$, $\tau_{\Phi}^{(0)}$, $\alpha_{\text{relax}}$, $\lambda_1, \lambda_2$, $T_{\max}$.

**Ensure:** $\tilde{\mathcal{G}} = (\tilde{\mathcal{V}}, \tilde{\mathcal{A}}, \tilde{\mathcal{W}})$

**Spectral Embedding:**

$$A_{ij} = \begin{cases} w_{ij}^0 \cdot e^{-d_{ij}/\gamma} \cdot \left(\frac{\phi_{ij}}{\max \phi}\right)^{\alpha}, & (i,j) \in \mathcal{A} \\ 0, & \text{otherwise} \end{cases}$$

$$\mathcal{L} = I - D^{-1/2} A D^{-1/2}, \quad D_{ii} = \sum_j A_{ij}$$

Compute first $m$ eigenvectors of $\mathcal{L}$ to obtain $\mathbf{X} \in \mathbb{R}^{|\mathcal{V}| \times m}$.

**Initialize:** $\tau_{\mathcal{E}} \leftarrow \tau_{\mathcal{E}}^{(0)}, \quad \tau_{\Phi} \leftarrow \tau_{\Phi}^{(0)}$

**for** $t = 1, \ldots, T_{\max}$ **do**

For each $v_i \in \mathcal{V}$:

$$r^* = \arg\min_{r \in \{1,\ldots,k\}} \left\{ \|\mathbf{x}_i - \mu_r\|^2 + \lambda_1 \cdot \mathbb{I}_{\text{DoSViol}} + \lambda_2 \cdot \mathbb{I}_{\text{DistViol}} \right\}$$

subject to:

$$\max_{v_j \in C_r} \left| w_{ij}^0 - \min_{v_\ell \in C_r} w_{i\ell}^0 \right| \le \epsilon_{\max}, \quad \max_{v_j \in C_r} d_{ij} \le \delta_{\max}$$

Assign $v_i \to C_{r^*}$ and update $\mu_r \leftarrow \frac{1}{|C_r|} \sum_{v_j \in C_r} \mathbf{x}_j$

Build $\tilde{\mathcal{G}}$ using definitions in Section II-C.

Compute:

$$\Delta\mathcal{E} = \left| \mathcal{E}^{\text{orig}} - \mathcal{E}^{\text{clust}} \right|, \quad \Delta\Phi = \left| \Phi^{\text{orig}} - \Phi^{\text{clust}} \right|$$

**if** $\Delta\mathcal{E} \le \tau_{\mathcal{E}}$ **and** $\Delta\Phi \le \tau_{\Phi}$ **then**

**return** $\tilde{\mathcal{G}}$

**else**

$\tau_{\mathcal{E}} \leftarrow \alpha_{\text{relax}} \cdot \tau_{\mathcal{E}}, \quad \tau_{\Phi} \leftarrow \alpha_{\text{relax}} \cdot \tau_{\Phi}$

**end if**

**end for**

**Fallback:** run unconstrained $k$-means on $\mathbf{X}$, build $\tilde{\mathcal{G}}$ from the resulting clusters, and **return** $\tilde{\mathcal{G}}$.

**Algorithm 2** GA for the original arc-level optimization problem

**Require:** Network $\mathcal{G}$, initial DoS vector $\mathbf{w}^0$, budget $B$, population size $N_{\text{pop}}$, maximum generations $N_{\text{gen}}$

**Ensure:** Approximate solution $\mathbf{w}^\star$

Initialize population $\mathcal{P}^{(0)} \subset \Omega$

**for** $n = 0, \ldots, N_{\text{gen}} - 1$ **do**

**for** each candidate $\mathbf{w}^{(j)} \in \mathcal{P}^{(n)}$ **do**

Recompute traffic factors, cyclist distances, and AON flows

Evaluate the exposure objective in Eq. (24)

Penalize or repair candidates violating Eq. (25)

**end for**

Select parents, apply crossover and mutation

Repair offspring if needed and form $\mathcal{P}^{(n+1)}$

**end for**

**return** feasible candidate with minimum exposure as $\mathbf{w}^\star$

**Algorithm 3** GA for the clustered optimization problem

**Require:** Clustered network $\tilde{\mathcal{G}}$, aggregated objects $\mathcal{G}_{\text{agg}}$, budget $B$, population size $N_{\text{pop}}$, maximum generations $N_{\text{gen}}$

**Ensure:** Approximate clustered solution $\tilde{\mathbf{w}}^\star$

Initialize population $\tilde{\mathcal{P}}^{(0)} \subset \tilde{\Omega}$

**for** $n = 0, \ldots, N_{\text{gen}} - 1$ **do**

**for** each candidate $\tilde{\mathbf{w}}^{(j)} \in \tilde{\mathcal{P}}^{(n)}$ **do**

Recompute clustered traffic factors, distances, and AON flows

Evaluate $\mathcal{E}^{\text{clust}}(\tilde{\mathbf{w}}^{(j)})$

Penalize or repair candidates violating Eq. (31)

**end for**

Select parents, apply crossover and mutation

Repair offspring if needed and form $\tilde{\mathcal{P}}^{(n+1)}$

**end for**

**return** feasible candidate with minimum clustered exposure as $\tilde{\mathbf{w}}^\star$

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**Algorithm 4** GW-based transfer and lifting of upgrade strategies

**Require:** Source solution $\mathbf{w}^{\star,\mathcal{H}}$, target network $\mathcal{G}$, clustered target $\tilde{\mathcal{G}}$, GW plan $\mathrm{t}^\star$, target budget $B^{\mathcal{G}}$
**Ensure:** Transferred strategy $\mathbf{w}^{\star,\mathrm{trans},\mathcal{G}}$

Compute source node scores $u_i^{\mathcal{H}}$
Transfer them to cluster scores $u_{C_r}^{\mathcal{H}\to\tilde{\mathcal{G}}}$ using $\mathrm{t}^\star$
Rank clustered objects using the transferred scores
Select feasible upgrades on $\tilde{\mathcal{G}}$ under budget $B^{\mathcal{G}}$
Denote the selected clustered vector by $\tilde{\mathbf{w}}^{\star,\mathrm{trans},\tilde{\mathcal{G}}}$
**for** each original arc $a \in \mathcal{A}^{\mathcal{G}}$ **do**
    Find $g$ such that $a \in \mathcal{A}^{\mathcal{G}}(g)$
    Set

$$w_a^{\star,\mathrm{trans},\mathcal{G}} = \max\left\{w_a^{\mathcal{G}}, \tilde{w}_g^{\star,\mathrm{trans},\tilde{\mathcal{G}}}\right\}$$

**end for**
Evaluate the lifted strategy on $\mathcal{G}$
**return** $\mathbf{w}^{\star,\mathrm{trans},\mathcal{G}}$


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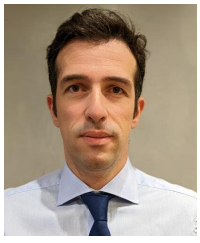
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**Manuel Campero Jurado** received the B.S. degree in Applied Mathematics in 2019 and the M.S. degree in Mathematics in 2021 from the Autonomous University of the State of Hidalgo, Mexico, and the Ph.D. degree in Automatic Control from Grenoble INP–Université Grenoble Alpes, France, in 2026. His interests include transport-network modeling, optimization, cycling safety, accessibility, and mobility systems.



**Carlos Canudas de Wit** received the B.S. degree in electronics and communications from the Monterrey Institute of Technology, Mexico, in 1980, and the M.S. and Ph.D. degrees in automatic control from the Grenoble Institute of Technology, France, in 1984 and 1987, respectively. He is a Directeur de recherche at CNRS, Grenoble, where he leads the NeCS Team at GIPSA-Lab and INRIA. He has served in several editorial and scientific leadership roles in the control community. His research focuses on electric vehicles, optimal control, traffic modeling, networked systems, and large-scale networks.



**Giovanni De Nunzio** received the B.S. and M.S. degrees in information and automation engineering from the University of L'Aquila, Italy, in 2007 and 2010, respectively, and the Ph.D. degree in automatic control from Grenoble INP, France, in 2015. He is a Research Engineer and Research and Innovation Project Leader with IFP Energies nouvelles, Lyon, France. His research focuses on transportation and mobility systems, driver-assistance systems, graph theory, optimization, and data science.